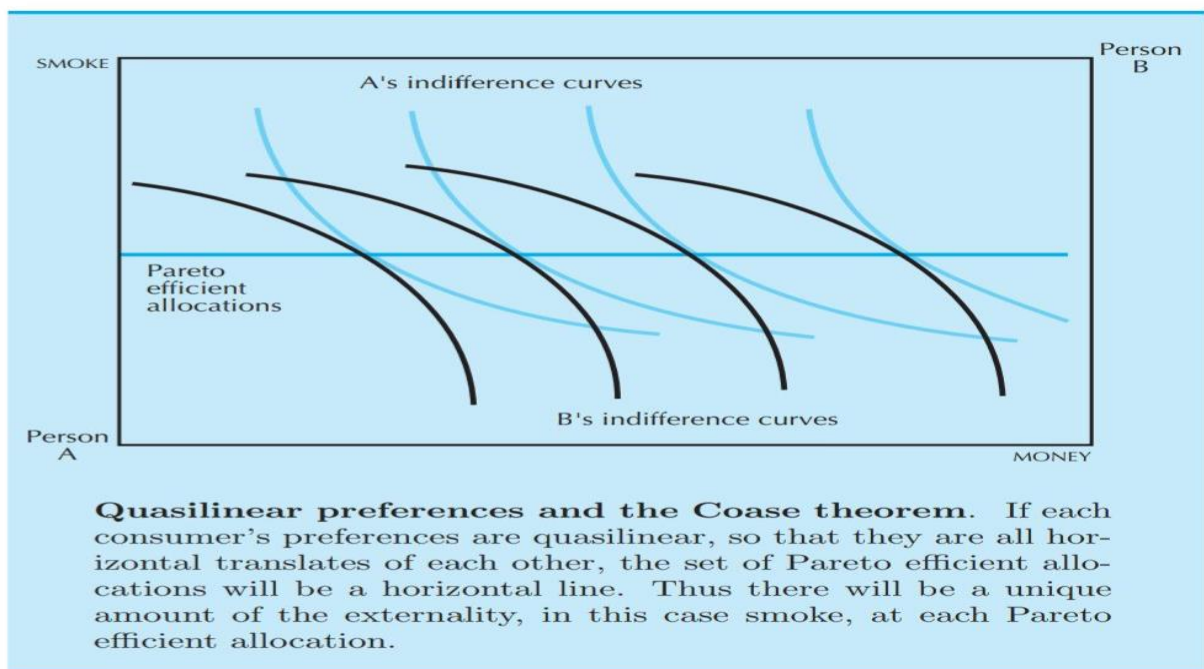


**In what situation the amount of smoke will remain fixed irrespective of nature of property recognised and enforced and pattern of distribution of income?**

**Answer:** As long as we have well-defined property rights in the good involving the externality—no matter who holds the property rights—the agents can trade from their initial endowment to a Pareto efficient allocation. If we want to set up a market in the externality to encourage trade, that will work as well. Problem arises if the property rights are not well defined. If A believes that he has the right to smoke and B believes that he has the right to clean air, we will face difficulties. The practical problems with externalities generally arise because of poorly defined property rights. Cases where property rights are poorly defined can lead to an inefficient production of externalities—which means that there would be ways to make both parties involved better off by changing the production of externalities. In the case of the two roommates, the amount of smoke generated will depend on whether the smoker has the property rights or the nonsmoker has them. But there is a special case where the outcome of the externality is independent of the assignment of property rights. If the agents' preferences are quasilinear, then every efficient solution must have the same amount of the externality. The amount of smoke is the same in every Pareto efficient allocation; only the dollar amounts held by the agents differ across the efficient allocations.



The result that under certain conditions the efficient amount of the good involved in the externality is independent of the distribution of property rights is sometimes known as the Coase Theorem. However, it should be emphasized just how special these conditions are. The quasilinear preference assumption implies that the demands for the good causing the externality doesn't depend on the distribution of income. Therefore a reallocation of endowments doesn't affect the efficient amount of the externalities. This is sometimes expressed by saying that the Coase theorem is valid if there are no "income effects."<sup>1</sup> In this case, the Pareto efficient allocations will involve a unique amount of the externality being generated. The different Pareto efficient allocations will involve different amounts of money being held by the consumers; but the amount of the externality—the amount of smoke—will be independent of the distribution of wealth.

So, we can say that if there is no well-defined property rights between two roommates, the amount of smoke will remain fixed irrespective of nature of property recognised and enforced and pattern of distribution of income.